

Animation

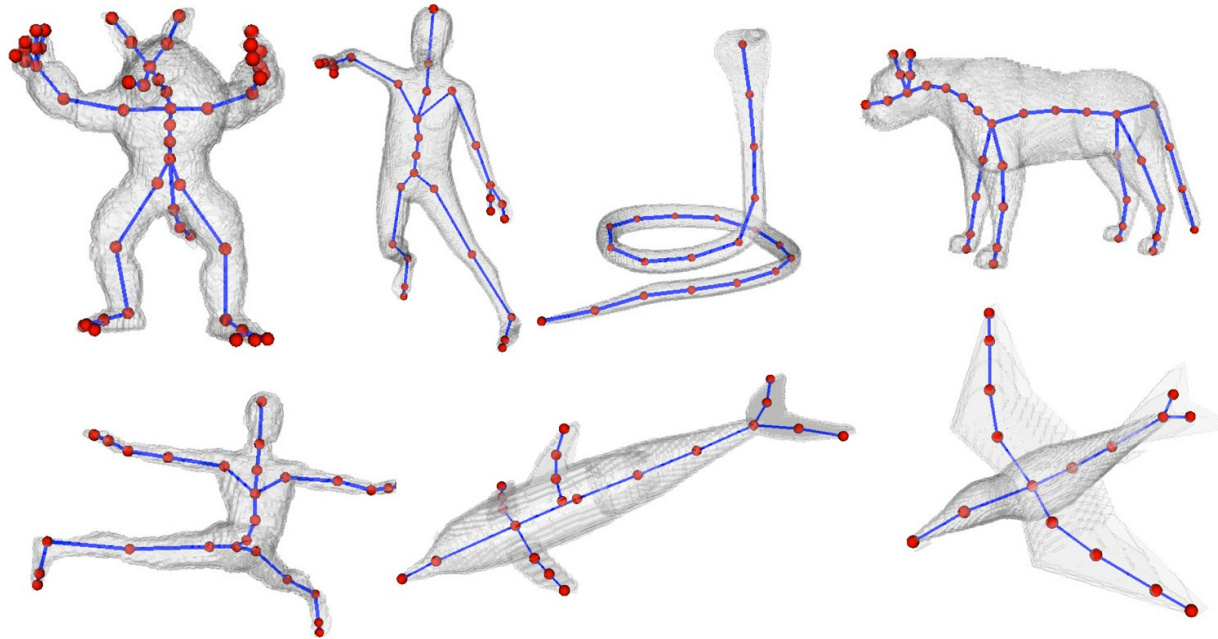
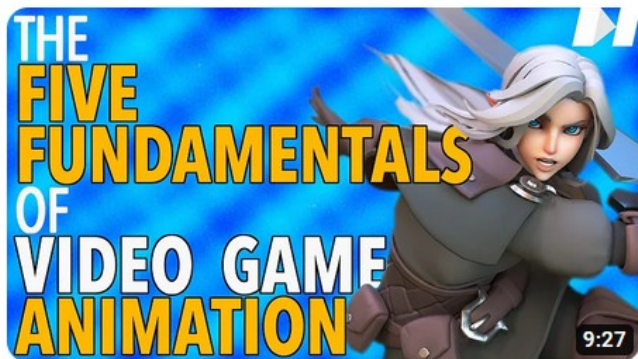


Image sources: <https://www.pinterest.com/pin/501166264768374898/>, <https://www.semanticscholar.org/paper/A-fully-automatic-rigging-algorithm-for-3D-Pantu Wong-Sugimoto/86e061ad83cab187a422b1e7119f915ed530533e>

Introduction



The Five Fundamentals of Game Animation: An Introduction

74K views • 3 years ago



Video Game Animation Study ✓

The 12 Principles of animation are the foundation for good animation, but video games have their own set of fundamentals ...

CC



It'll Be Alright - YouTube Royalty Free Music | See You Soon - YouTube Royalty Free Music |...

3 chapters ▾

<https://www.youtube.com/watch?v=CrOGr8i575k>



- Characters need to move in a fluid, organic way
 - Much more challenging to animate than rigid objects like vehicles, projectiles, soccer balls, and Tetris pieces, ...
- The character animation system imbues characters with natural-looking motion
 - Any game object that is not 100% rigid can take advantage of the animation system.
 - Examples
 - Vehicle with moving parts
 - A piece of articulated machinery
 - Trees waving gently in the breeze
 - Exploding building
 - ...
 - Chances are good that these objects makes at least partial use of the game engine's animation system

Cel Animation



- Known as traditional animation, or hand-drawn animation
- Used in the earliest animated cartoons
- The illusion of motion is produced by displaying a sequence of still pictures known as frames in rapid succession
- Cel animation is a specific type of traditional animation

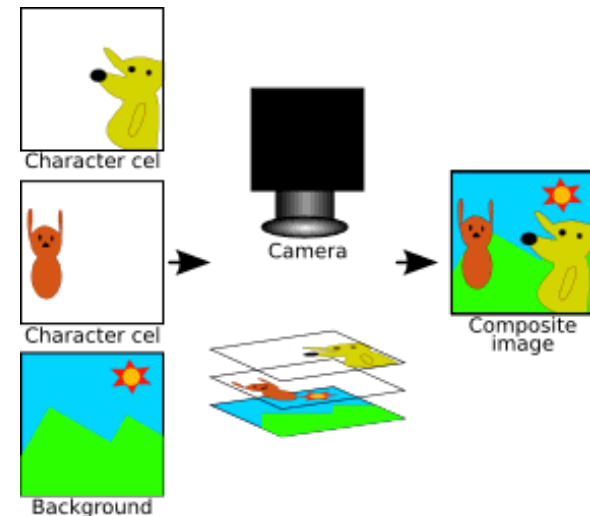
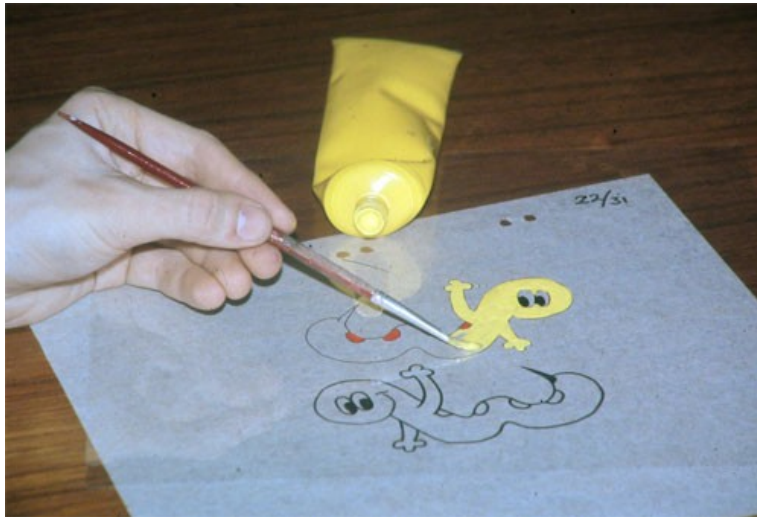


Image sources: <https://conceptartempire.com/cel-animation/>, https://en.wikipedia.org/wiki/Traditional_animation

Sprite animation



- Sprite animation is the electronic equivalent to cel animation
- A sprite is a small bitmap that can be overlaid on top of a full-screen background image without disrupting it
- Often drawn with the aid of specialized graphics hardware
- A sequence of frames designed so that it animates smoothly even when it is repeated indefinitely is known as a looping animation
- Example: run cycle (used in multiple Intellivision games)



- Characters typically have a number of looping animation cycles, including various idle cycles, a walk cycle, and a run cycle

Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Sprite Animation Examples (1/2) – Mega Man

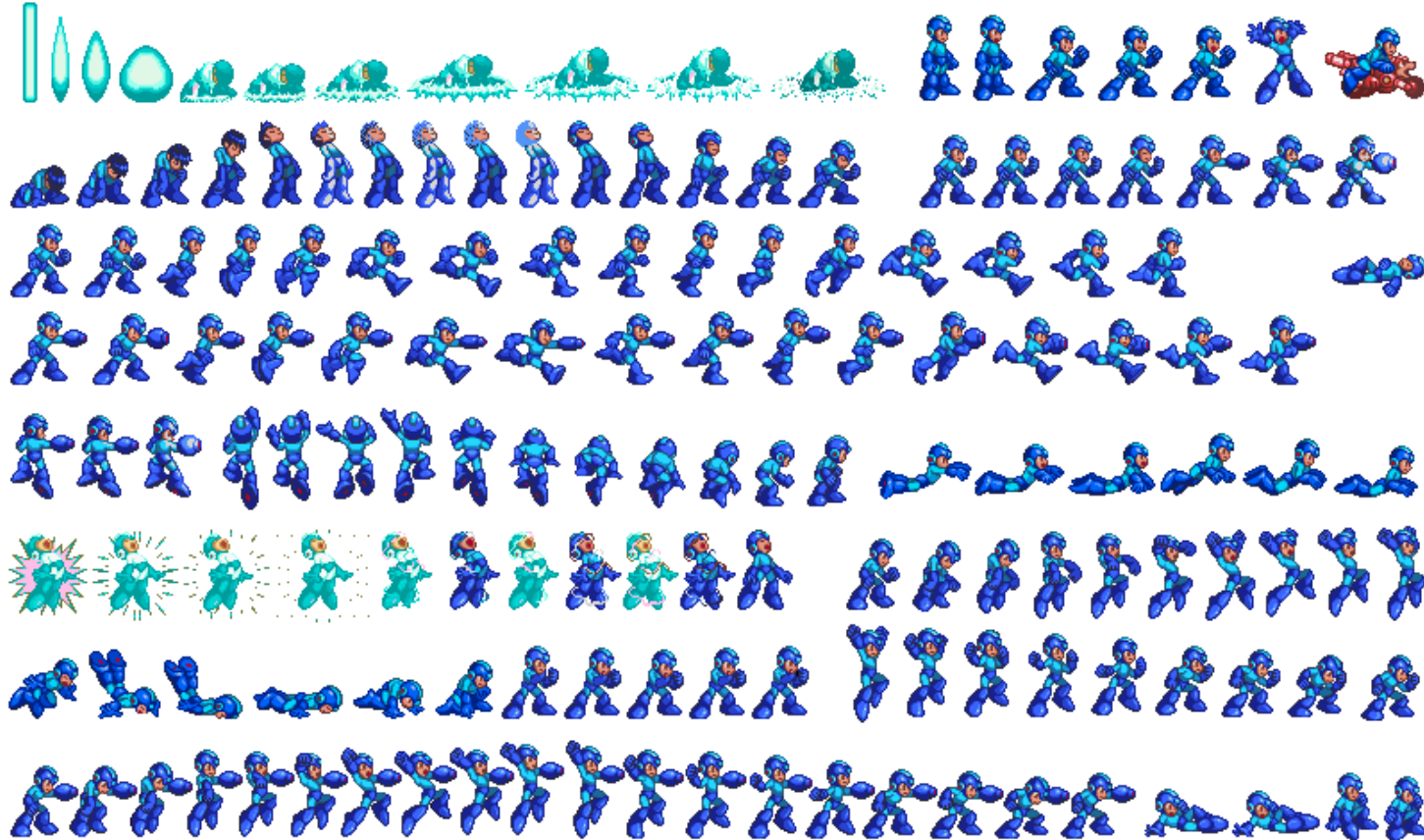


Image sources: <https://www.pinterest.com/pin/563231497122914794/>

Sprite Animation Examples (2/2) – ALTTP



Image sources: <https://spritedatabase.net/file/8705/Link>

Pros and Cons of Sprite Animation



- Pros
 - Hand-drawn sprites have a certain charm and appeal
 - Relatively easy to create
 - Can be drawn by hand
 - Can be rendered (3ds Max, Maya, ...)
 - It's also easy to add additional elements (fire, blanket)
 - Requires only a very simple animation system
- Cons
 - No blending or fading between different animations
 - No scaling of animation speeds (1 animation frame per game frame)
 - Animations can't be mixed together (head, torso, legs, ...)
 - Storage requirements can become very high
 - Dependence on the perspective (walking up or down)
 - Asymmetric sprites are twice as complex

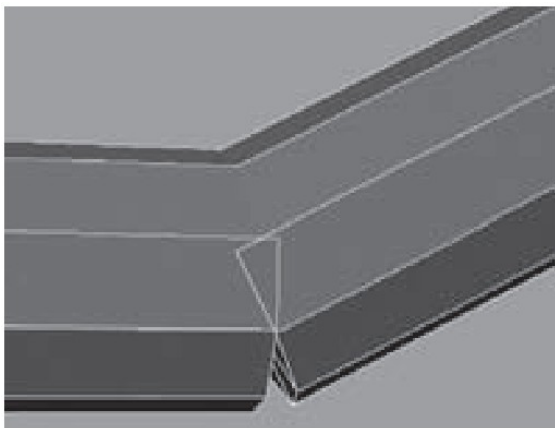


Image sources: <https://jasonpictures.deviantart.com/art/Death-Battle-Bio-Peacock-531320030>, <https://giphy.com/stickers/sprites-haggar-scrollboss-hG4DZEz2noAtW>

Rigid Hierarchical Animation



- The earliest approach to 3D character animation
 - A character is modelled as a collection of rigid pieces
 - The rigid pieces are constrained to one another in a hierarchical fashion
 - Is often not very pleasing due to “cracking” at the joints
 - Good for robots and machinery that really is constructed of rigid parts



Pelvis

Torso

UpperRightArm

LowerRightArm

RightHand

UpperLeftArm

UpperLeftArm

LeftHand

Head

UpperRightLeg

~~LowerRightLeg~~

RightFoot⁹

Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014, <https://minecraft-nl.gamepedia.com/Minecraft.net>

Per-Vertex Animation and Morph Targets



- Per-Vertex Animation
 - The vertices of the mesh are animated, and motion data is exported
 - Can produce any mesh deformation imaginable
 - Very data-intensive technique, since time-varying motion information must be stored for each vertex of the mesh
- Morph Targets
 - Vertices of a mesh are moved by an animator to create a relatively small set of fixed, extreme poses
 - Animations are produced by blending between two or more of these fixed poses at runtime (simple linear interpolation → LERP)
 - The morph target technique is often used for facial animation
 - Morph target animation gives an animator full control over every vertex of a facial mesh

Morph Targets Example

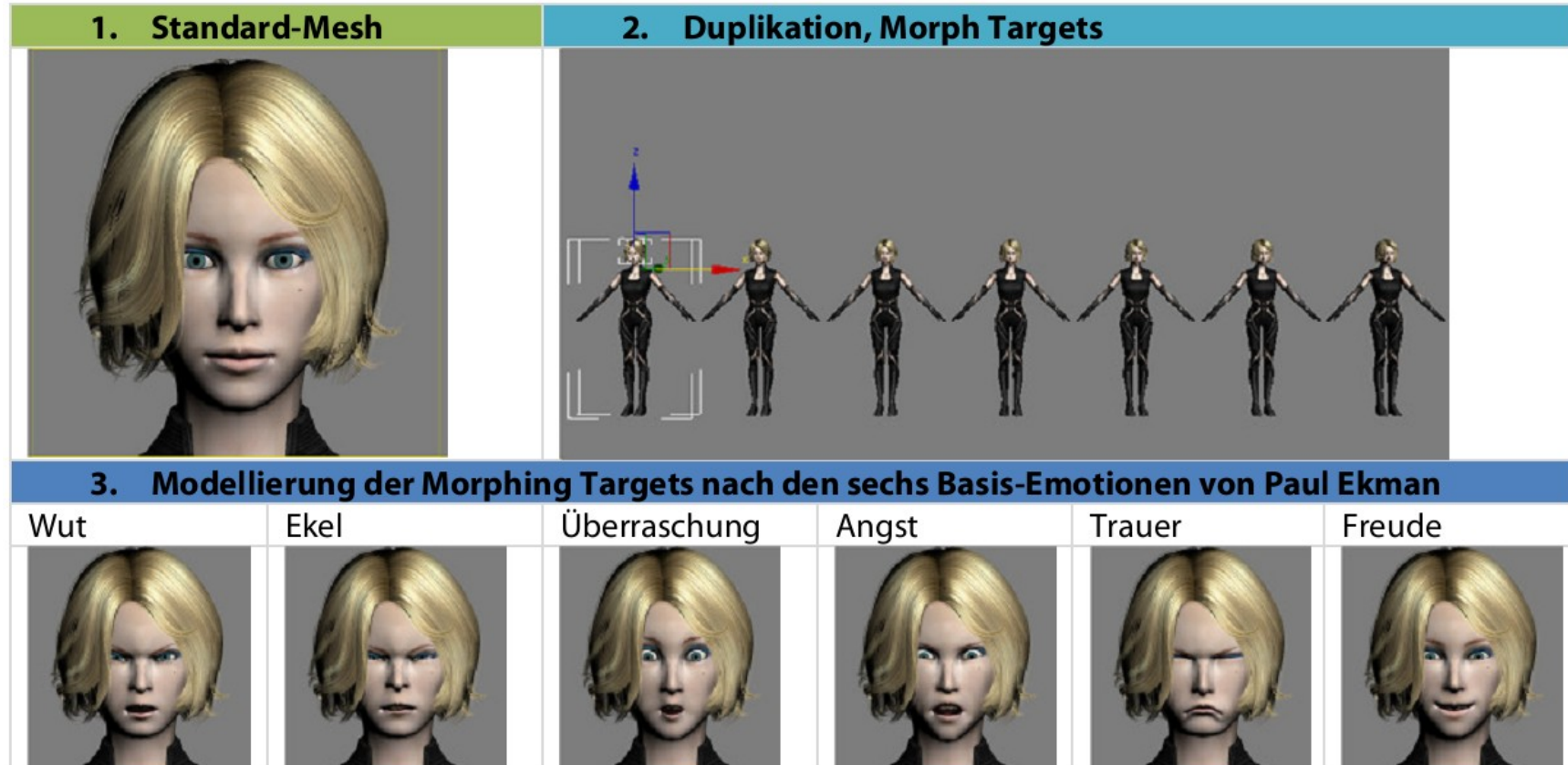


Image sources: Solveigh Jäger, “Die erfolgreiche Konstruktion virtueller Spielfiguren in 3D-Computerspielen”, 2012



- This technique has many of the benefits of per-vertex and morph target animation
 - Permits the triangles of an animated mesh to deform
 - More efficient performance and memory usage characteristics
 - Produces reasonably realistic approximations to the movement of skin and clothing
- In skinned animation, a skeleton is constructed from rigid bones, just as in rigid hierarchical animation.
 - The rigid pieces are not rendered on-screen, they remain hidden
- A smooth continuous triangle mesh called a skin is bound to the joints of the skeleton
 - Its vertices track the movements of the joints
 - Each vertex of the skin mesh can be weighted to multiple joints
 - The skin can stretch in a natural way as the joints move

Skinned Animation Example

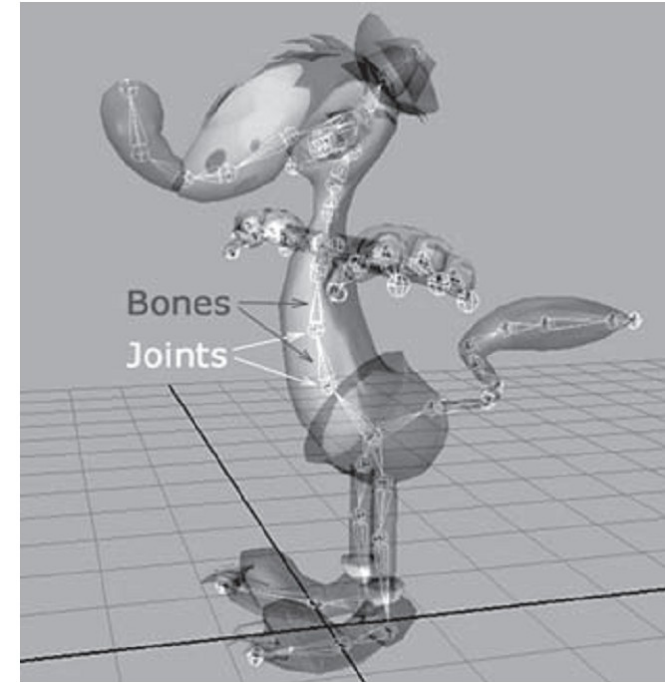


Image sources: Solveigh Jäger, "Die erfolgreiche Konstruktion virtueller Spielfiguren in 3D-Computerspielen", 2012 Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Skeletons



- A skeleton is comprised of a hierarchy of rigid pieces known as joints
 - The terms joint and bone are often used interchangeably
 - Joints are the objects that are directly manipulated by the animator
 - The bones are simply the empty spaces between the joints

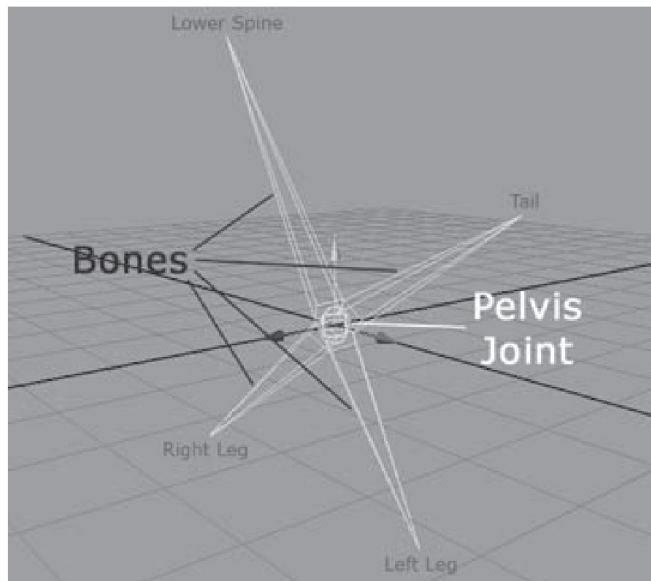
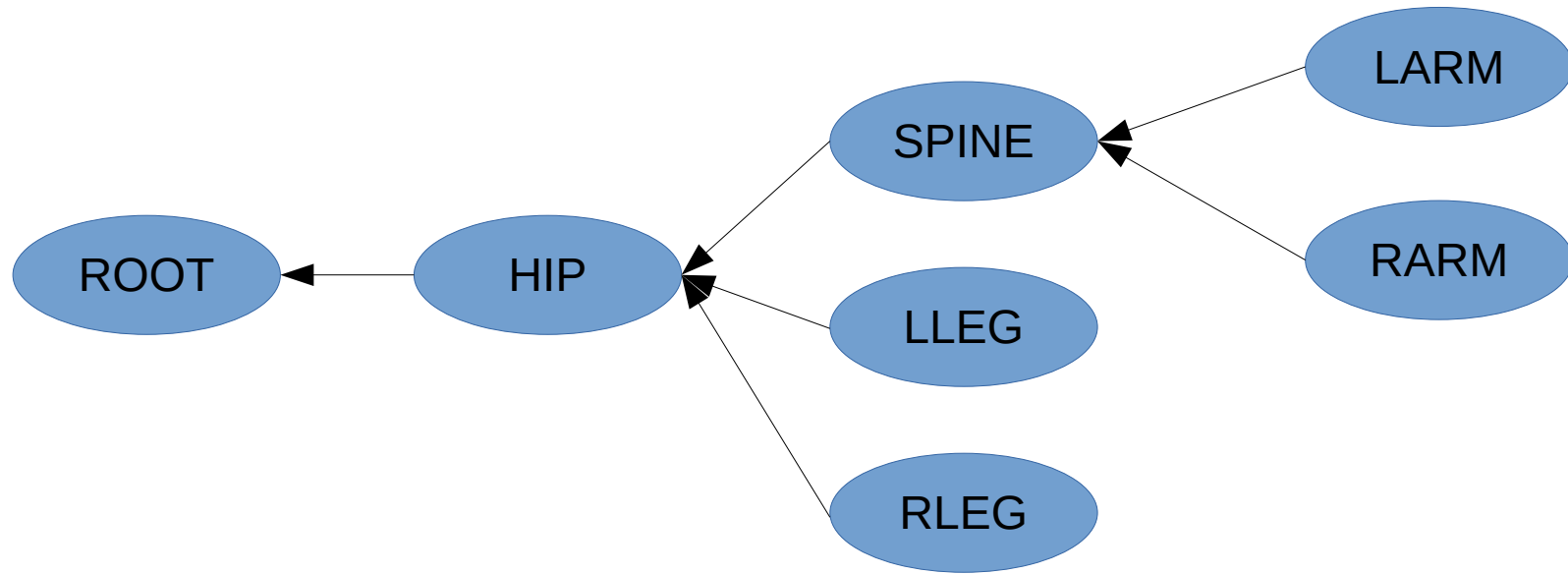


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

The Skeletal Hierarchy



- The joints in a skeleton form a hierarchy or tree structure
 - One joint is selected as the root, and all other joints are its children, grandchildren, and so on
 - Each joint has one and only one parent
 - A skeleton can be fully described by storing the index or name of its parent with each joint





- Every animation takes place over time
 - Discrete, still poses, displayed in rapid succession
 - Usually at a rate of 30 or 60 poses per second
- In skeletal animation, the pose of the skeleton directly controls the vertices of the mesh
 - Posing is the animator's primary tool for breathing life into her characters
 - A skeleton is posed by rotating, translating, and possibly scaling its joints
 - The pose of a joint is defined as the joint's position, orientation, and scale, relative to some frame of reference
 - The pose of a skeleton is just the set of all of its joints' poses and is normally represented as a simple array of matrices or SQTs.

Bind Pose



- Bind Pose (also called the reference pose or the rest pose)
 - The pose of the 3D mesh prior to being bound to the skeleton
 - Also called the T-pose because the character is usually standing with his feet slightly apart and his arms out-stretched in the shape of the letter T

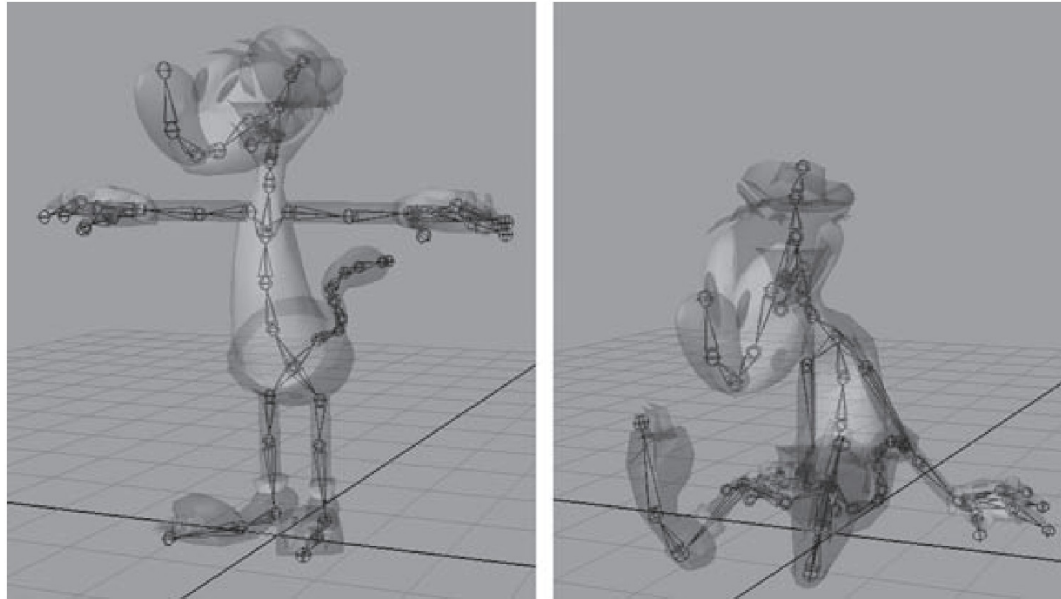


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Local Poses



- A joint's pose is most often specified relative to its parent joint.
 - Allows a joint to move naturally (E.g. arm follows shoulder)
- The term local pose describes a parent-relative pose
 - Every joint defines a set of local coordinate space axes
- Mathematically, a joint pose is nothing more than an affine transformation
 - The pose of an entire skeleton P_{skel} can be written as the set of all poses P_j , where j ranges from 0 to $N-1$:

$$\mathbf{P}^{\text{skel}} = \left\{ \mathbf{P}_j \right\}_{j=0}^{N-1}$$

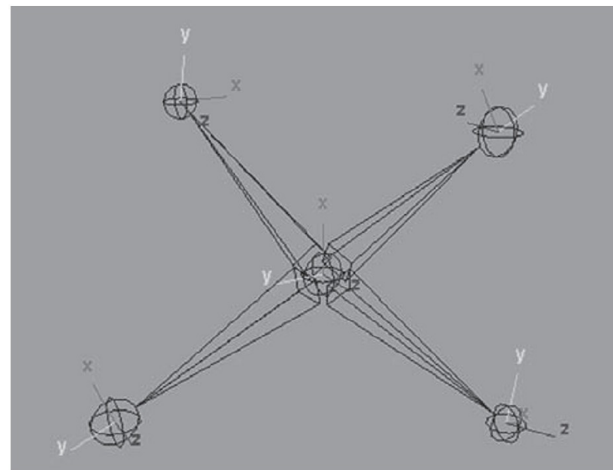
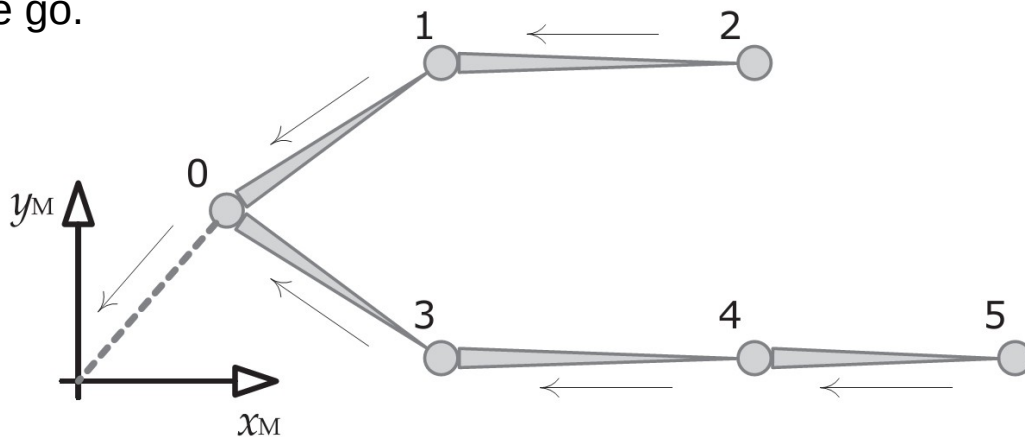


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Global Poses



- A joint's pose in model space or world space is called a global pose
 - Mathematically, the model-space pose of a joint ($j \rightarrow M$) can be found by walking the skeletal hierarchy from the joint in question all the way to the root, multiplying the local poses ($j \rightarrow p(j)$) as we go.



- Why do we need global poses?
 - Hit detection, parenting, camera target, ...

Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Clips (1/2)



- A game character's movement must be broken down into a large number of fine-grained motions.
 - Player input is unpredictable!
- We call these individual motions animation clips, or sometimes just animations.
 - Each clip causes the character to perform a single well-defined action
 - Some clips are designed to be looped
 - Walk cycle, run cycle or idle cycle, ...
 - Others clips are designed to be played once
 - Throwing an object, or tripping and falling to the ground, ...
 - Some clips affect the entire body of the character
 - Jumping into the air, swinging a sword, ...
 - Other clips affect only a part of the body
 - Waving an arm, turning the head, ...

Clips (2/2)



- The movements of any one game character are typically broken down into literally thousands of clips.
- Exceptions to this rule include
 - In-game cinematic (IGC)
 - Non-interactive sequence (NIS)
 - Full-motion video (FMV)
 - These are created in much the same way as computer-generated films
 - Each scene is planned out before any animations are created
 - The entire scene can be animated as one long, contiguous sequence of frames
- Variations to this rule
 - Semi-interactive sequences known as a quick time events (QTEs)
 - Several clips are used, but they are much longer
 - Or a sequence of different NISs is shown

The Local Time Line (1/4)



- The rate at which frames are displayed is not necessarily the same as the rate at which poses are created by the animator
 - The animator generates important poses known as key poses or key frames at specific times within the clip
 - The computer calculates the poses in between via linear or curve-based interpolation
 - We can actually sample the pose of the character at any time
 - In other words, an animation clip's time line is continuous

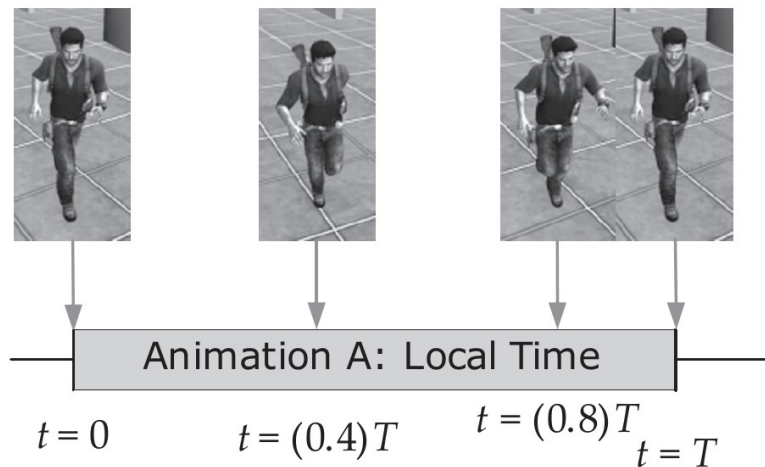


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

The Local Time Line (2/4)



- In computer animation, the time variable t is a real (floating-point) number, not an integer.
 - Real-time game's frame rate always varies a little
 - Game animations are sometimes time-scaled (faster, slower, reverse)
 - An animation clip is almost never sampled on integer frame numbers
- In game animation, time is both continuous and scalable

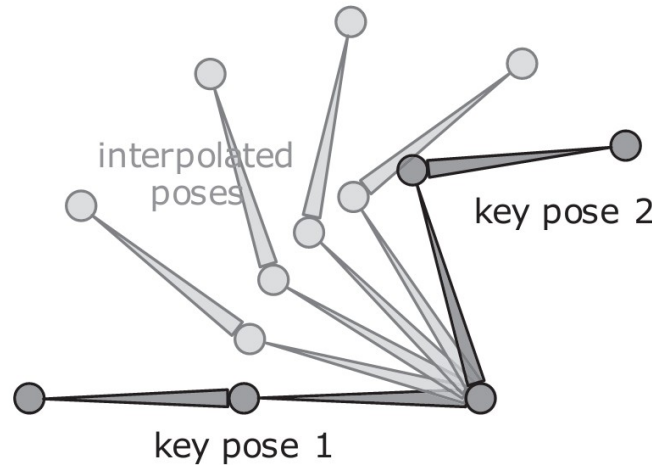
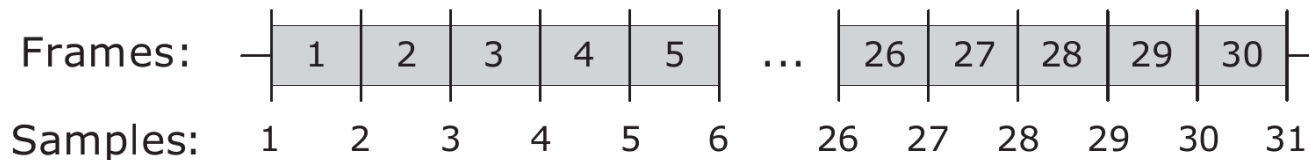


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

The Local Time Line (3/4)



- Sample → A single point in time (signal processing)
- Frame → A time period that is 1/30 or 1/60 of a second in duration



- If a clip is non-looping, an N-frame animation will have N+1 unique samples
- If a clip is looping, then the last sample is redundant, so an N-frame animation will have N unique samples

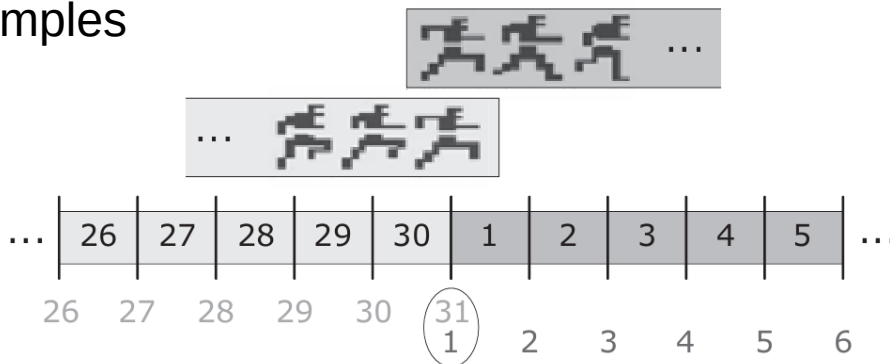


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

The Local Time Line (4/4)



- Phase → Normalized time of the animation
 - Using a normalized time unit u , such that $u=0$ at the start of the animation, and $u=1$ at the end, no matter what its duration T may be
 - Useful when synchronizing two or more animation clips
 - E.g. cross-fading between two walk cycles with different length
 - Also easier to communicate and to comprehend
 - E.g. the hit sound must be played at 60% of the animation's local time

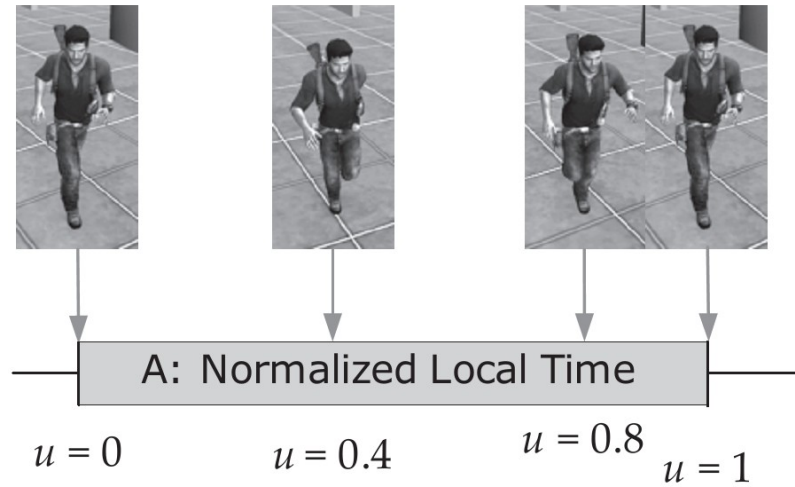
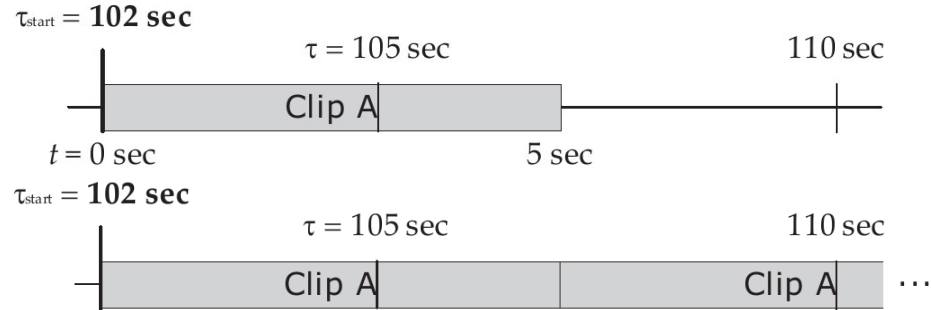


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

The Global Time Line



- We can think of playing an animation as simply mapping that clip's local time line onto the character's global time line



- Time-scaling a clip makes it appear to play back more quickly or more slowly than originally animated.

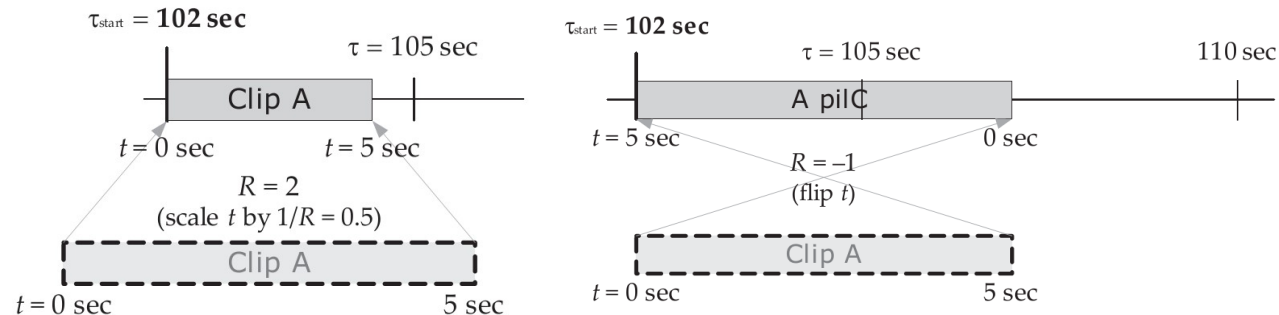


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Skinning



- The process of attaching the vertices of a 3D mesh to a posed skeleton is known as skinning
 - Each vertex can be bound to one or more joints
 - If bound to a single joint, the vertex tracks that joint's movement exactly
 - If bound to two or more joints, the vertex's position is a weighted average
 - Per-Vertex Skinning Information
 - The index or indices of the joint(s) to which it is bound
 - A weighting factor for each joint to which it is bound

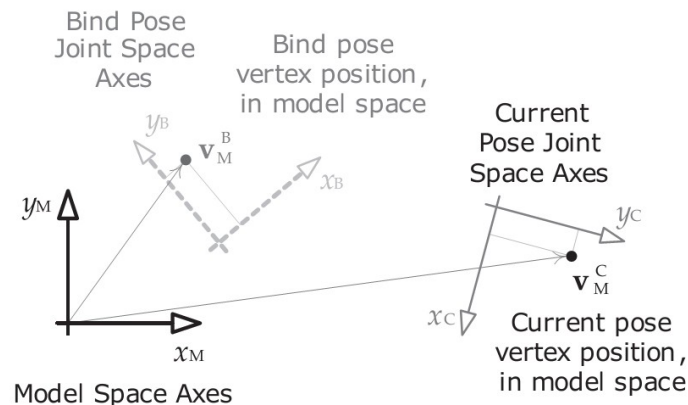
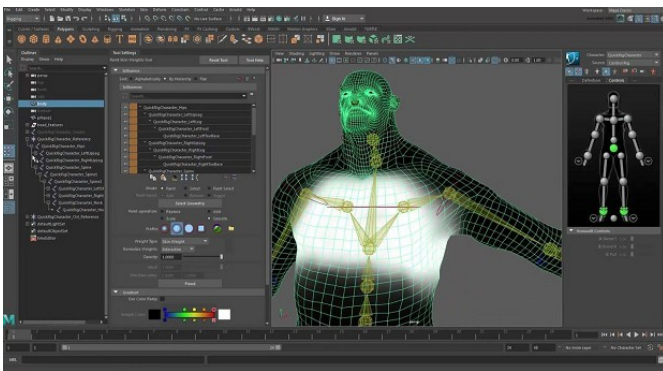


Image sources: <https://www.pinterest.com/pin/363736107390461406/>, Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Animation Blending (1/5)



- The term animation blending refers to any technique that allows more than one animation clip to contribute the final pose of the character.
 - Blending combines two or more input poses to produce an output pose
- Two or more poses at a single point in time
 - Combine two or more animations into a host of new animations
 - Walk + injured walk, aim left + aim right, ...
- Find an intermediate pose between two known poses at different points in time
- Smooth transition from one animation to another
 - Gradually blending from the source animation to the destination over a short period of time

Animation Blending (2/5)



- Cross-blended transition
 - Smooth transition
 - Time lines must be synchronized so that the positions of the legs and arms in one clip match up roughly with their positions in the other clip
 - Frozen transition
 - Works well when the two clips are unrelated and cannot be time-synchronized

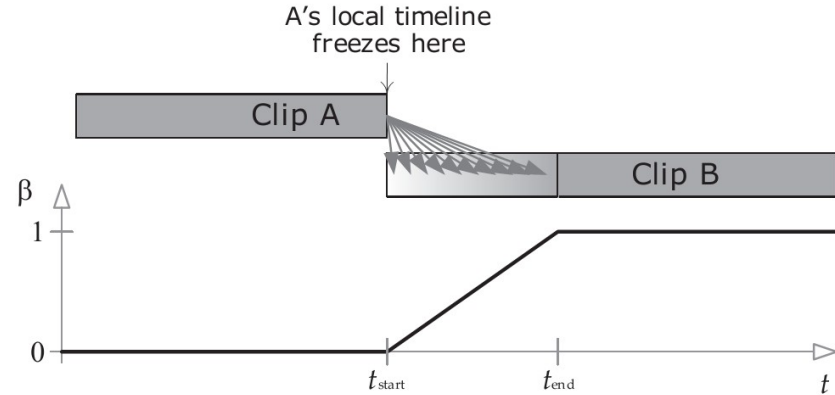
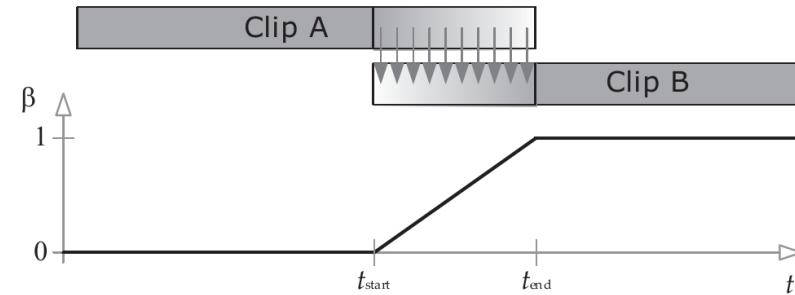


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Animation Blending (3/5)



- Partial-Skeleton Blending
 - A human being can control different parts of his or her body independently
 - Blend percentage varies on a per-joint basis
 - Has a tendency to make a character's movement look unnatural
 - Can cause the movements of one part of the body to appear disconnected from the rest of the body
 - The movements of a real human body are never totally independent



Image sources: <https://www.amazon.de/Music-Party-Rock-House-Wii/dp/B004JMZPM8>, <http://www.intercambiosvirtuales.org/juegos/winter-sports-2012-feel-the-spirit-pc-game>

Animation Blending (4/5)



- Additive Blending

- Basic Idea

- Source clip (S)
 - Reference clip (R)
 - Difference clip (D) = $S - R$

- Example

- S = running
 - R = running in a tired manner
 - D = $S - R$ = tired
 - T = walking
 - $\Rightarrow T + D$ = walking in a tired manner

- Limitations

- Tends to over-rotate the joints
 - We must be careful when selecting R and T
 - Different animations for each pose

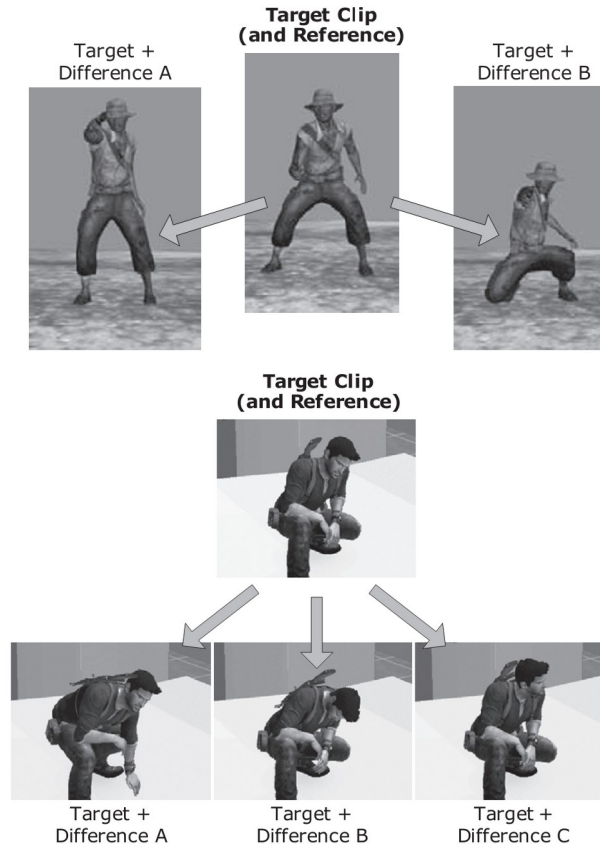


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Animation Blending (5/5)



- Aim and Look-At
 - Standard animation (walking, running, ...)
 - Plus at least two extreme poses (look 100% left and 100% right)
 - The angle of the aim is governed by the additive blend factor of each clip

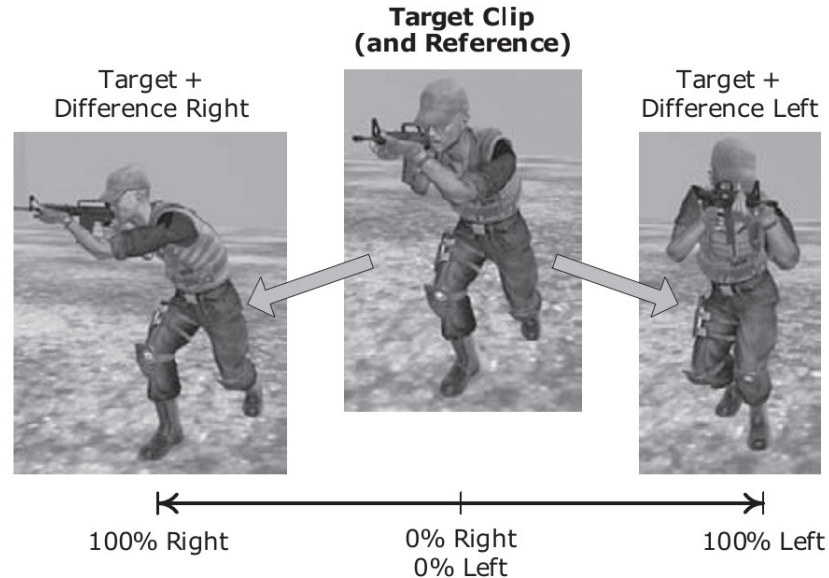


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Animation Post-Processing



- It is often necessary to modify the pose prior to rendering the character (animation post-processing)
- Procedural Animations
 - Any animation generated at runtime
 - Tire rotation, steering wheel, tree movement, ...
 - These adjustments are made to the local pose
- Inverse Kinematics
 - The input is the desired global pose of a single joint, which is known as the end effector
 - Error minimization problem: There might be one solution, many, or none at all

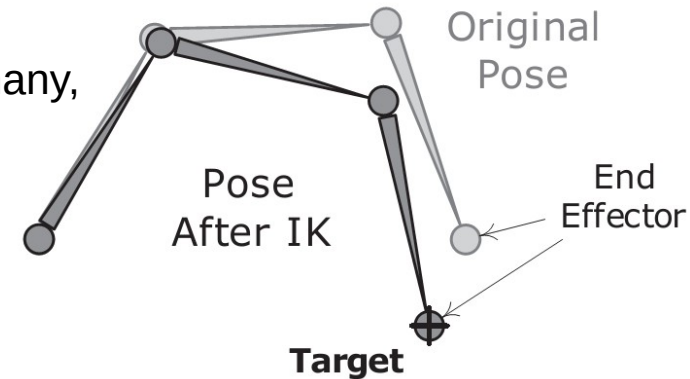


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Data Structures (1/2)

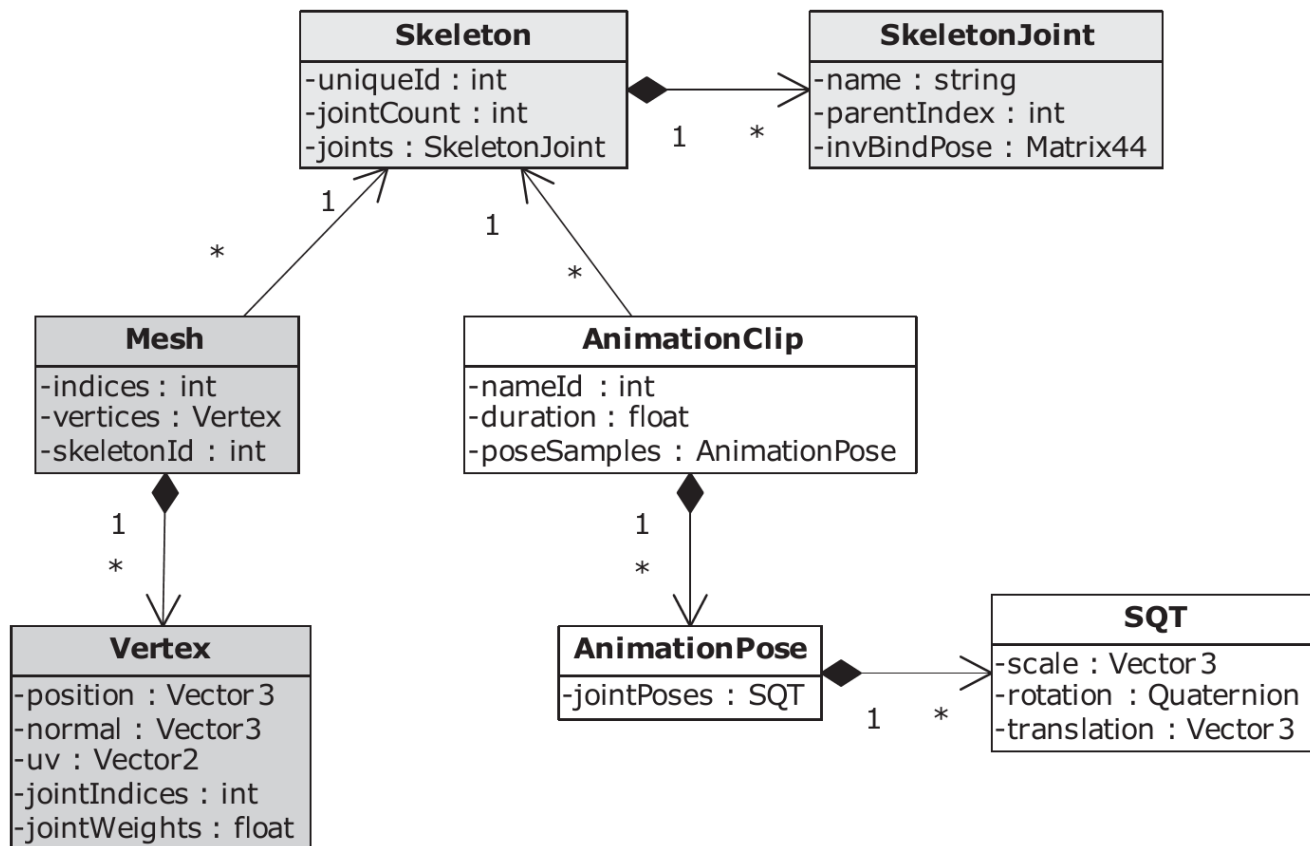


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Data Structures (1/2)



- Usually only a few skeletons with many different skins
 - Game designers often try to reduce the number of unique skeletons in the game to one, or just a few, because each new skeleton generally requires a whole new set of animation clips.

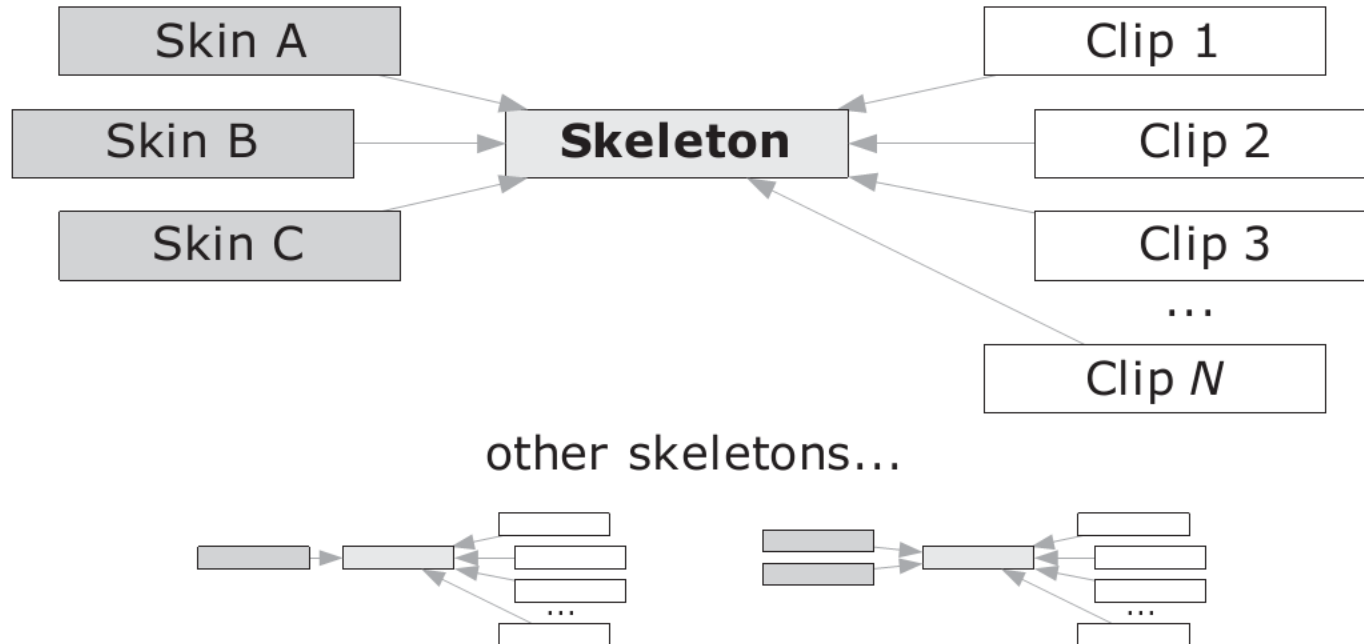


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014

Compound Objects (1/4)



- Model space / object space / local space
 - When a triangle mesh is created the positions of the triangles' vertices are specified relative to a Cartesian coordinate system
 - The model space origin is placed at a central location within the object
 - Center of mass
 - Ground between the feet of a humanoid or animal character
 - Most game objects have an inherent directionality
 - Up, front, left right

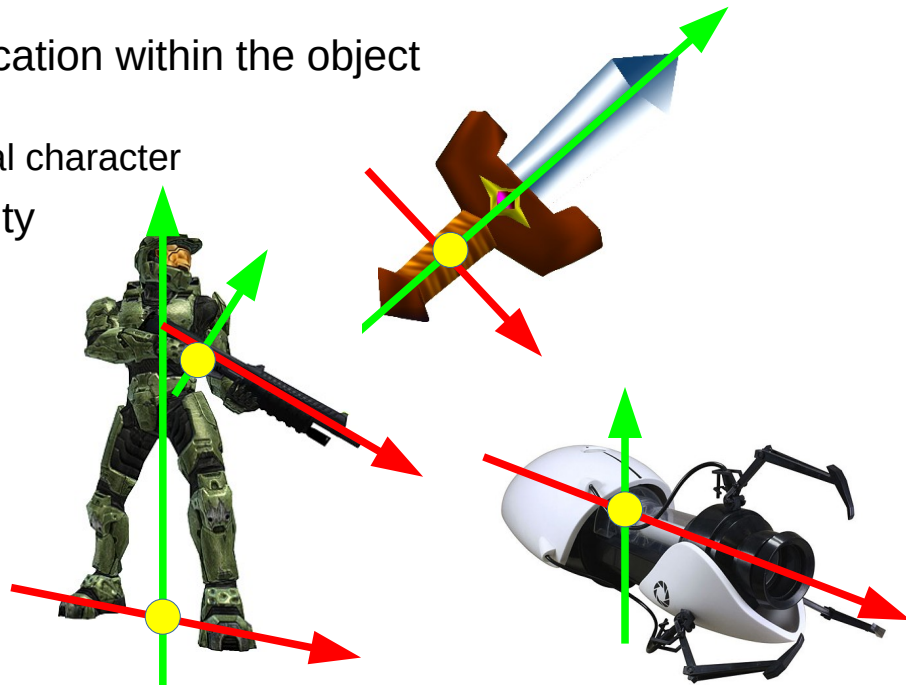
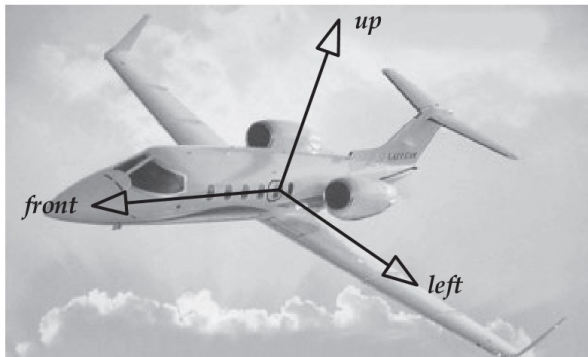


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014, <https://toraiinexamikaze.deviantart.com/art/Halo-2-Master-Chiet-251551553>, <https://www.planetminecraft.com/project/minecraft-antigravity-portal-gun-test-chambers-plus-mods/>, [http://zelda.wikia.com/wiki/File:Kokiri_Sword_\(Ocarina_of_Time\).png](http://zelda.wikia.com/wiki/File:Kokiri_Sword_(Ocarina_of_Time).png)

Compound Objects (2/4)



- World space is a fixed coordinate space, in which the positions, orientations, and scales of all objects in the game world are expressed.
 - Location of the world-space origin is arbitrary
 - The orientation of the x-, y-, and z-axes is arbitrary

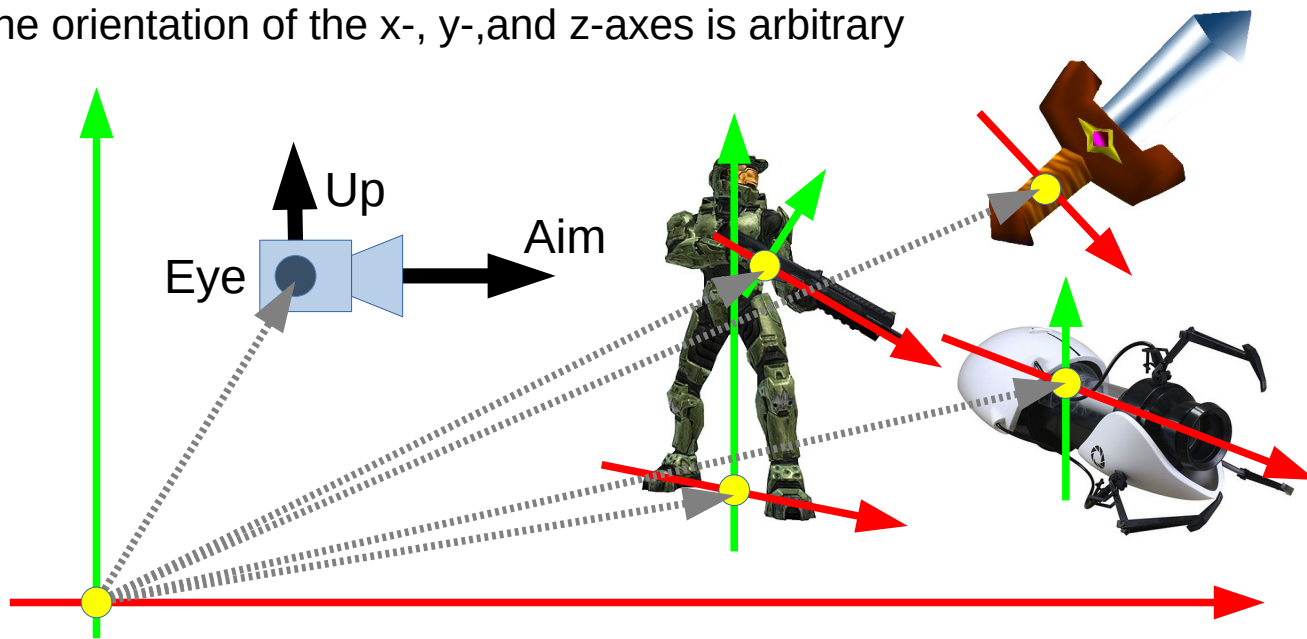


Image sources: Game Engine Architecture, 2nd edition, Taylor & Francis Ltd., 2014, <https://toraiinexamikaze.deviantart.com/art/Halo-2-Master-Chief-251551553>, <https://www.planetminecraft.com/project/minecraft-antigravity-portal-gun-test-chambers-plus-mods/>, [http://zelda.wikia.com/wiki/File:Kokiri_Sword_\(Ocarina_of_Time\).png](http://zelda.wikia.com/wiki/File:Kokiri_Sword_(Ocarina_of_Time).png)

Compound Objects (3/4)



- Linking objects to joints
 - Parent → World space transformation of a joint
 - Child → Center of the local coordinate system

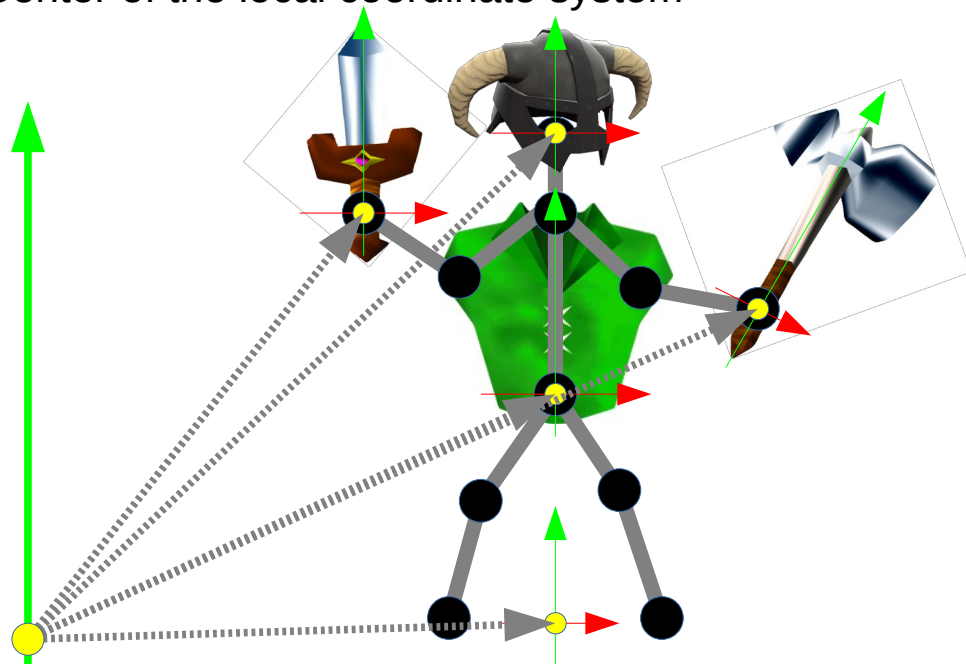


Image sources: [http://zelda.wikia.com/wiki/File:Kokiri_Sword_\(Ocarina_of_Time\).png](http://zelda.wikia.com/wiki/File:Kokiri_Sword_(Ocarina_of_Time).png), http://zelda.wikia.com/wiki/Megaton_Hammer, http://zelda.wikia.com/wiki/Kokiri_Tunic, <https://www.giantbomb.com/dragonborn-helmet/3055-5389/characters/>

Compound Objects (4/4)



- Examples

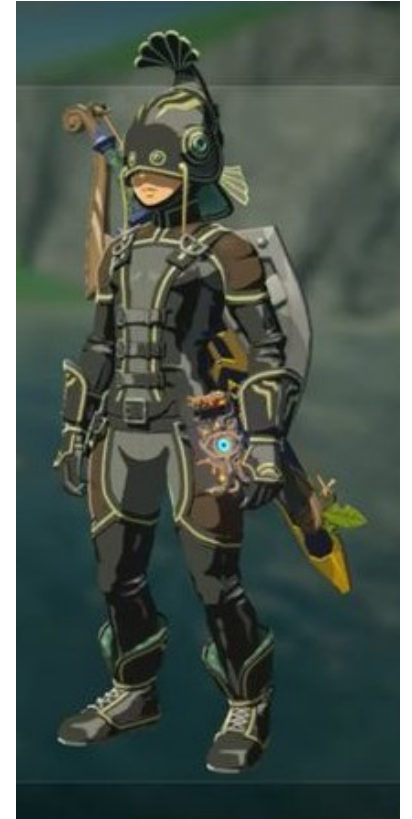
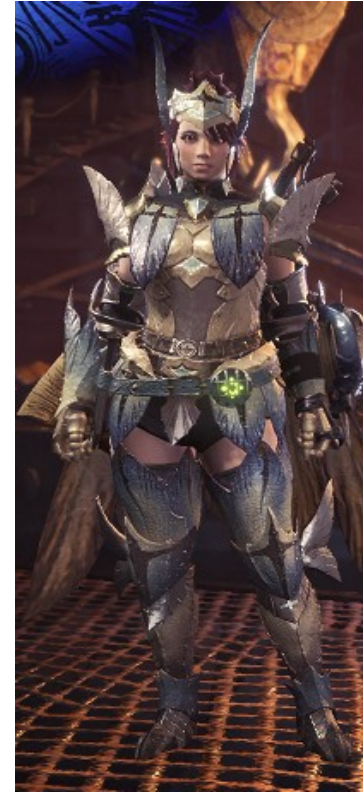


Image sources: <https://www.gamezgeneration.de/winter-sports-2012-feel-the-spirit-3/>, <https://www.game.co.uk/en/super-mario-galaxy-2-39186>, <https://mic.com/articles/171256/zelda-breath-of-the-wild-rubber-armor-how-to-get-the-shock-resistant-gear>, <https://www.usgamer.net/articles/12-04-2019-monster-hunter-world-beginners-guide/best-armor-set-guide>, https://cdn.cheatcc.com/images/imagesps3/littlebigplanetkarting_5a.jpg

In-game Cutsscenes



- What else would we need to be able to create in-game cinematics?
 - Timer (or some other kind of event generator)
 - E.g.: Musical Instrument Digital Interface (MIDI)
 - Timing is very accurate (right on the beat)
 - Note values indicate different kinds of events
 - Script system
 - Defined set of actions or freely programmable
 - Reactions to individual events
 - Start or stop animations
 - Camera cut, fade or movement
 - Play sound effects or music
 - Show or hide subtitles
 - Start or stop particle emitters
 - Show or hide objects
 - Change to a different scene
 - Send character from A to B
 - ... anything!

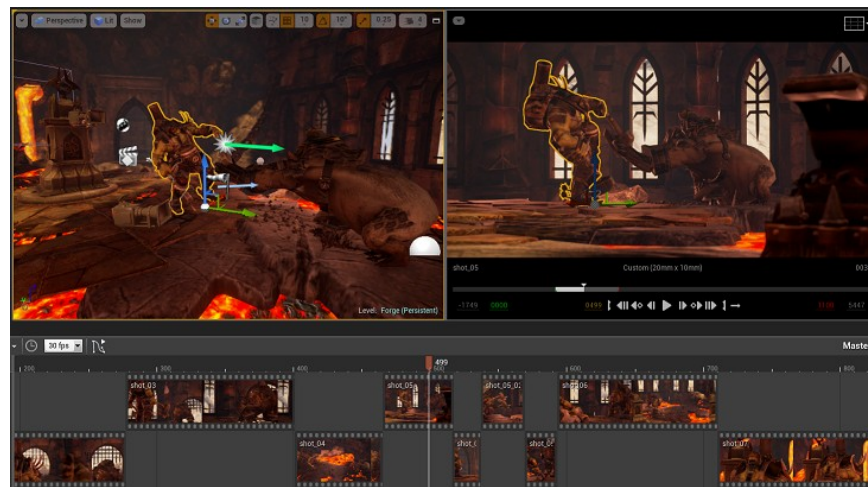


Image sources: <https://docs.unrealengine.com/en-US/Engine/Sequencer/index.html>